Alex Rivera

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Professional Summary

Game Engine Programmer with 5+ years building core engine systems for AAA and indie titles. Expert in C++, graphics programming, engine architecture, and performance optimization. Shipped 3 commercial titles across PC, PlayStation, and Xbox. Passionate about rendering pipelines, physics systems, and tools development.

Technical Skills

C++, C#, HLSL, GLSL, DirectX 11/12, Vulkan, OpenGL, Unity, Unreal Engine 4/5, PhysX, Havok, FMOD, Wwise, CMake, Perforce, Git, Visual Studio, RenderDoc, PIX, Nsight, Python, Lua, CMake, Premake, Jenkins, GitLab CI, Agile, Scrum, Jira, Confluence

Professional Experience

Senior Engine Programmer - Apex Games Studio

2021-Present

- Lead core engine systems for unannounced AAA open-world title on Unreal Engine 5
- Architected and implemented asynchronous asset streaming system reducing load times by 60%
- Developed custom render graph framework supporting deferred, forward+, and hybrid rendering paths
- Optimized GPU performance achieving stable 60 FPS at 4K on PS5/Xbox Series X
- Mentored 3 junior engineers and established engine coding standards

Engine Programmer - Nebula Interactive

2018-2021

- Built core runtime systems for proprietary C++ game engine (Nebula Engine)
- Implemented physically-based rendering pipeline with image-based lighting and BRDF
- Developed animation system with inverse kinematics, blend trees, and motion matching
- Integrated PhysX physics and created custom character controller with ragdoll
- Built editor tools in C#/WPF for level design, material graph, and animation state machines
- Ported engine from DirectX 11 to DirectX 12 and Vulkan for cross-platform support

Gameplay Programmer - IndieForge Games

2016-2018

- Implemented gameplay systems for 2 shipped indie titles (Unity/C#)
- Designed ability system with data-driven configuration for 100+ abilities
- Built networking layer for co-op multiplayer using Photon and custom prediction
- Created procedural generation systems for dungeon crawler title

Education

MSc Computer Science (Graphics & Simulation), University of Surrey (2016)
BSc Computer Games Programming, University of Hertfordshire (2014)

Key Projects

- Nebula Engine - Custom C++ game engine with PBR renderer, physics, audio, and editor
- Voxel Engine - Procedural terrain with marching cubes, LOD, and GPU instancing
- Path Tracer - Physically-based path tracer with BVH acceleration and importance sampling